

Name: AVIRAJ KUMAR RAI AVIRAJ KUMAR RAI | Reg. No: RA2511003011026

Unit 5 : Session 5 : SLO 2 | Polynomial and Rational Expression Operations

I. Questions & Answers

- Multiply $(x+2)(x+3)$ → expand $((x + 2)*(x + 3))$ → $x^2 + 5x + 6$.
- Factor $x^2 - 4$ → factor $(x^2 - 4)$ → $(x - 2)(x + 2)$ (difference of squares).
- Add $1/(x+1) + 1/(x-1)$ → together/simplify: combine over the common denominator $(x+1)(x-1)$ → $(x - 1) + (x + 1) / (x^2 - 1) = 2x/(x^2 - 1)$.

II. Code

- from sympy import symbols, expand, factor, simplify
- x = symbols('x')
- print(expand((x + 2)*(x + 3))) # $x^2 + 5x + 6$
- print(factor(x**2 - 4)) # $(x - 2)(x + 2)$
- print(simplify(1/(x + 1) + 1/(x - 1))) # $2x/(x^2 - 1)$